ISLAMIC UNIVERSITY, KUSHTIA, BANGLADESH
Department of Information and Communication Technology
ICT-4202: Artificial Intelligence and Machine Learning Laboratory
FINAL LABORATORY EXAMINATION - SET 1

Instructions: Write, execute and demonstrate all required programs/tasks. Save source code/notebook/WEKA output and explain the result during viva.

Q1. Intelligent Agent and PEAS [25 Marks]

Develop a Python program for a two-room Simple Reflex Vacuum Agent. Initially: Room A = Dirty, Room B = Clean, Initial location = B. Include Suck, Left, Right and NoOp. Modify the agent so that it stops when both rooms are clean. Display every percept-action pair. [15]
Write the PEAS description for an autonomous robot that delivers documents between offices of a university building. [10]

Q2. Search Algorithms [30 Marks]

Use graph: A:{B,C}, B:{D,E}, C:{F}, D:{G}, E:{G}, F:{G}, G:{}. Implement BFS, DFS and Greedy Best-First Search. Use h(A)=6, h(B)=4, h(C)=3, h(D)=2, h(E)=2, h(F)=1, h(G)=0. Find a path from A to G. Display the path and nodes expanded by each algorithm. Compare the results.

Q3. Machine Learning - k-NN [35 Marks]

Load the Iris dataset. Use a 70/30 stratified train/test split with random_state=42. Train k-NN for k=1,3,5,7. Report test accuracy for each and identify the best k. Display the confusion matrix. Standardize the features and repeat the best k-NN experiment. Comment on the effect of scaling.

Q4. Viva Voce [10 Marks]

Agent, PEAS, BFS/DFS/Greedy, heuristic, k-NN, scaling and confusion matrix.

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FINAL LABORATORY EXAMINATION - SET 2

Instructions: Write, execute and demonstrate all required programs/tasks. Save source code/notebook/WEKA output and explain the result during viva.

Q1. Knowledge-Based Agent [25 Marks]

Construct a knowledge base: Cloudy; Cloudy->Rain; Rain->WetGround; WetGround->Slippery; Slippery->DriveSlowly; Rain AND Cold->WearJacket. Write a Python forward-chaining algorithm. Run it without Cold and with Cold. Display initial facts, rule-firing sequence and final inferred knowledge. Explain why WearJacket does or does not fire.

Q2. DLS and IDDFS [30 Marks]

Use: S->{A,B}, A->{C,D}, B->{E,F}, C->{H}, D->{G}, E->{I}, F->{J}. Implement DLS with limit=2, DLS with limit=3 and IDDFS. Search from S to G. Display result and trace. Explain cutoff and repeated shallow search.

Q3. Linear Regression [35 Marks]

Use data: (1,4), (2,5), (3,8), (4,9), (5,11), (6,14). Fit linear regression; print slope and intercept; predict score for 7 study hours; calculate R^2; plot observed data and fitted line. Change the score at 6 hours from 14 to 18, refit, compare R^2 and explain the change.

Q4. Viva Voce [10 Marks]

Forward chaining, facts/rules, DLS, IDDFS, slope, intercept and R^2.

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FINAL LABORATORY EXAMINATION - SET 3

Instructions: Write, execute and demonstrate all required programs/tasks. Save source code/notebook/WEKA output and explain the result during viva.

Q1. First-Order Logic Representation [25 Marks]

Represent: Parent(Ali,Babu), Parent(Ali,Rina), Parent(Babu,Karim), Parent(Rina,Sumi), Parent(Karim,Nila). Implement functions for children, grandparent relations and sibling relations. Display all inferred relations. Explain tuple matching for Parent(x,y) AND Parent(y,z) -> Grandparent(x,z).

Q2. A* Search [30 Marks]

Weighted edges: S-A=2, S-B=4, A-C=2, A-D=5, B-D=1, C-G=5, D-G=3. Heuristic: h(S)=6, h(A)=5, h(B)=4, h(C)=4, h(D)=2, h(G)=0. Implement A*. Report expansion sequence, g(n), h(n), f(n), optimal path and total cost. Then set every h(n)=0, rerun and explain.

Q3. K-Means Clustering [35 Marks]

Use the first two Iris features. Run K-means for k=2,3,4,5. Record inertia and silhouette score for each k. Select a suitable k. Plot clusters and centroids. Explain why cluster label 0 is not automatically Iris class 0.

Q4. Viva Voce [10 Marks]

Predicate, constants, variables, A*, g/h/f, admissibility, centroid, inertia and silhouette.

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FINAL LABORATORY EXAMINATION - SET 4

Instructions: Write, execute and demonstrate all required programs/tasks. Save source code/notebook/WEKA output and explain the result during viva.

Q1. Search Comparison [30 Marks]

Use graph: P->{Q,R}, Q->{S,T}, R->{U}, S->{V}, T->{V}, U->{V}, V:{}. Implement BFS, DFS and Bidirectional Search from P to V. Print path, expansion order and number of expanded nodes. Reverse the adjacency order of Q's children and determine whether DFS changes.

Q2. WEKA Classification [30 Marks]

Using iris.arff in WEKA Explorer: inspect the dataset and set the correct class attribute. Run J48 with 10-fold cross-validation. Run IBk with k=1,3,5. Record accuracy and confusion matrix; compare J48 with the best IBk result; save screenshots/result-buffer output.

Q3. Linear Regression [30 Marks]

Use X=[2,4,6,8,10,12] and Y=[5,9,13,17,21,25]. Fit linear regression; calculate slope, intercept, prediction at X=15 and R^2; plot data and fitted line. Modify one target value and discuss the effect on R^2.

Q4. Viva Voce [10 Marks]

BFS vs DFS, bidirectional search, cross-validation, J48, IBk and regression.

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FINAL LABORATORY EXAMINATION - SET 5

Instructions: Write, execute and demonstrate all required programs/tasks. Save source code/notebook/WEKA output and explain the result during viva.

Q1. Intelligent Agent [25 Marks]

Design a three-location cleaning agent with locations A, B and C. Each location may be Clean or Dirty. The agent must clean the current location if dirty, otherwise move to another location, and use NoOp when all are clean. Display location, percept, action and environment state after each step. Also provide a PEAS description.

Q2. Greedy and A* Search [30 Marks]

Edges: A-B=2, A-C=3, B-D=4, B-E=2, C-E=2, D-G=3, E-G=4. Heuristic: h(A)=6, h(B)=5, h(C)=4, h(D)=2, h(E)=3, h(G)=0. Implement Greedy Best-First and A*. Print expansion order, path and final path cost. Explain why smallest h(n) alone may not produce the lowest-cost path.

Q3. Decision Tree Classification [35 Marks]

Using Iris, create a 70/30 stratified split. Train Decision Trees with max_depth=1,2,3,4,None. Report train and test accuracy. Display the confusion matrix for the best model; discuss underfitting/overfitting; report precision, recall and F1-score.

Q4. Viva Voce [10 Marks]

Simple reflex agent, state, Greedy vs A*, tree splitting, overfitting and classification metrics.

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FINAL LABORATORY EXAMINATION - SET 6

Instructions: Write, execute and demonstrate all required programs/tasks. Save source code/notebook/WEKA output and explain the result during viva.

Q1. Propositional Logic [25 Marks]

Use: Registered; Registered->Student; Student->HasID; Student AND PaidFee->ExamEligible; ExamEligible->CanEnterExamHall. Develop forward chaining and test: Registered only, and Registered plus PaidFee. Display inference trace and final knowledge. Also construct the truth table of P->Q and identify when it is false.

Q2. BFS, DFS and IDDFS [30 Marks]

Use: S->{A,B}, A->{C,D}, B->{E,F}, C->{G}, D->{H}, E->{I}, F->{J}. Search S to G using BFS, DFS and IDDFS. Print path and expanded nodes. Compare search behavior and memory characteristics.

Q3. k-NN vs Decision Tree [35 Marks]

Using Iris, use the same 70/30 stratified split. Train k-NN with k=5 and Decision Tree with max_depth=3. Compute accuracy and confusion matrices. Identify misclassified samples. Standardize features and rerun k-NN. Explain why scaling affects k-NN more than a decision tree.

Q4. Viva Voce [10 Marks]

Forward chaining, implication, BFS/DFS/IDDFS, k-NN and decision tree.

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FINAL LABORATORY EXAMINATION - SET 7

Instructions: Write, execute and demonstrate all required programs/tasks. Save source code/notebook/WEKA output and explain the result during viva.

Q1. FOL and Knowledge Representation [25 Marks]

Use: Parent(Rahman,Hasan), Parent(Rahman,Nabila), Parent(Hasan,Rafi), Parent(Nabila,Tania), Parent(Rafi,Samia). Implement children_of(x), grandchildren_of(x), grandparent(x,z) and siblings(x,y). Avoid duplicates and self-sibling relations.

Q2. Greedy Best-First Search [30 Marks]

Graph: S->{A,B,C}, A->{D}, B->{E}, C->{F}, D->{G}, E->{G}, F->{G}. Heuristic: S=7, A=4, B=5, C=3, D=2, E=2, F=1, G=0. Implement Greedy Best-First Search. Then modify heuristics so the algorithm explores a less useful branch first. Show both traces and explain.

Q3. K-Means and Feature Scaling [35 Marks]

Using all four Iris numeric features, run K-means with k=3 without scaling. Record centers, inertia and silhouette score. Standardize all features and repeat. Compare results. Visualize any two features. Explain why centroids and inertia cannot be compared naively across differently scaled spaces.

Q4. Viva Voce [10 Marks]

FOL, relation, heuristic, Greedy Best-First, unsupervised learning, scaling and silhouette.

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FINAL LABORATORY EXAMINATION - SET 8

Instructions: Write, execute and demonstrate all required programs/tasks. Save source code/notebook/WEKA output and explain the result during viva.

Q1. DLS and IDDFS [25 Marks]

Use: A->{B,C}, B->{D,E}, C->{F}, D->{H}, E->{I}, F->{G}. Implement DLS with depth 2, DLS with depth 3 and IDDFS to search for G. Display result, limit, trace at each IDDFS iteration and number of times root A is visited. Explain the trade-off of repeated shallow searches.

Q2. WEKA Experiment [30 Marks]

Using Iris in WEKA, inspect all attributes and run IBk k=5 using 10-fold cross-validation. Apply normalization/standardization and repeat IBk. Then run J48. Record accuracy and confusion matrices; compare and state which configuration performed best.

Q3. Regression Programming [35 Marks]

Create at least 10 numerical input-output observations for a reasonable prediction problem. Split into training/testing sets; fit Linear Regression; report slope/intercept where applicable; calculate train and test R^2. Make two predictions for unseen inputs; plot observations/model output; interpret the result and state one limitation.

Q4. Viva Voce [10 Marks]

DLS, IDDFS, cutoff, cross-validation, preprocessing, regression and R^2.

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FINAL LABORATORY EXAMINATION - SET 9

Instructions: Write, execute and demonstrate all required programs/tasks. Save source code/notebook/WEKA output and explain the result during viva.

Q1. BFS and DFS Comparison [30 Marks]

Graph: A->{B,C}, B->{D,E}, C->{F}, D->{H}, E->{G}, F->{G}, G:{}, H->{G}. Implement BFS and DFS from A to G. Print path and expansion trace. Change adjacency ordering and run both again. Explain which is affected more visibly and whether BFS returns a shallowest path in this unweighted graph.

Q2. A* and Heuristic Admissibility [30 Marks]

Edges: S-A=1, S-B=5, A-C=2, A-D=4, B-D=1, C-G=5, D-G=2. Choose and list an admissible heuristic. Implement A* and display g, h, f, optimal path and total cost. Then make one heuristic overestimate the true remaining cost. Discuss why admissibility matters.

Q3. K-Means Model Selection [30 Marks]

Using the first two Iris features, run K-means for k=2,3,4,5. Record inertia and silhouette. Plot an elbow graph and silhouette values against k. Select a suitable k; visualize final clusters and centers; explain why inertia usually decreases as k increases.

Q4. Viva Voce [10 Marks]

BFS/DFS, admissible heuristic, A*, inertia, silhouette and clustering.

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FINAL LABORATORY EXAMINATION - SET 10

Instructions: Write, execute and demonstrate all required programs/tasks. Save source code/notebook/WEKA output and explain the result during viva.

Q1. Integrated Knowledge-Based Agent [25 Marks]

Use: TemperatureHigh; TemperatureHigh->Hot; Hot AND Humid->Uncomfortable; Uncomfortable->FanOn; Rain->Humid. Implement forward chaining for: TemperatureHigh only; TemperatureHigh and Humid; TemperatureHigh and Rain. Print inference trace and final knowledge. Distinguish given and inferred facts.

Q2. Integrated Search Task [35 Marks]

Use structure: S->{A,B}, A->{C,D}, B->{D,E}, C->{G}, D->{G}, E->{G}. Assign positive edge costs and reasonable heuristic values. Implement BFS, Greedy Best-First and A*. Report expansion order, path and path cost. Discuss why shortest in number of edges and lowest-cost path need not be identical.

Q3. Classification [30 Marks]

Using Iris, make a stratified 70/30 split. Train k-NN with k=3 and Decision Tree with max_depth=3. Calculate accuracy, confusion matrices, precision, recall and F1-score. Compare errors and state which model you prefer for this run and why.

Q4. Viva Voce [10 Marks]

Knowledge base, forward chaining, BFS/Greedy/A*, classification metrics and model comparison.